

Device Compliance Registry

Overview

An AI-assisted, human-in-the-loop device registration triage prototype

DEMONSTRATION

The National Device Assurance Agency, the manufacturers, and the device data shown here are illustrative. This is a demonstration prototype, not a production system or a live service.

Document type: **Overview**

Status: **Interactive prototype.**

1. Introduction

The Device Compliance Registry is an interactive prototype showing how artificial intelligence can improve a regulatory workflow while keeping humans in authority over consequential decisions. It models the National Device Assurance Agency, an illustrative regulator responsible for registering grid-connected consumer energy devices such as home battery inverters, hybrid inverters, and EV chargers before they enter the market.

The prototype demonstrates a core principle for AI in regulated settings: the AI recommends and the human decides. Every automated step is transparent, every consequential decision is attributable to a named person, and the entire process is captured in an append-only audit trail.

Purpose

The prototype makes three things tangible:

- **Efficiency.** Routine, low-risk applications can be cleared automatically, freeing assessors to focus on cases that need judgement.
- **Safety and oversight.** Anything uncertain, novel, or carrying a safety flag is routed to a human, and the AI can never issue a final approval on its own.
- **Transparency and traceability.** The reasoning behind every routing decision is visible on screen, and every action is logged with its author.

2. What It Does

A manufacturer submits a device model for registration. On submission, the application flows through four stages:

- 1. AI triage.** An AI step assesses the application for risk and returns a structured recommendation: a risk score and band, a confidence value, a recommendation, plain-language reasoning, and a set of flags that each name the specific field they are based on.
- 2. Transparent routing rules.** A set of rules decides routing. It auto-approves only when the AI recommends approval and confidence is high and risk is low and there are no critical flags. Anything else is routed to a human.
- 3. Human review.** A human assessor reviews each routed case with the AI recommendation, reasoning, and the exact routing rules visible, then approves, rejects, or returns it for more information. The human decides and the AI only recommends.
- 4. Audit.** Every step is written to an append-only audit trail, attributing each action to a system process, a named model, or a named human, with a UTC timestamp.

THE CENTRAL SAFEGUARD

The confidence-and-risk routing rules are within the application logic, not in the model. They are visible, testable, and deliberately stricter than the model's recommendation. This is a defence-in-depth design so the routing decision never depends on the AI alone.

3. The Three Screens

The application is organised into three tabs, reflecting the two user roles and the oversight view.

Screen	Role	What it provides
Register a device	Manufacturer	A submission form for device and documentation details. Includes a one-click button to load five sample applications that demonstrate the full range of outcomes.
Assessor queue	Assessor	The human review workspace. Each routed case opens an assessment panel showing the AI recommendation, its reasoning and flags, and the pass or fail detail of every routing check. The assessor approves, rejects, or returns for information.
Assurance dashboard	Oversight	Operational metrics, including the routing split between auto-approved and human-routed cases, and the full append-only audit trail.

4. How the Triage Works

The triage assesses each application against several factors: certificate presence and expiry, safety attestation, test report presence, device novelty, manufacturer track record, and completeness of declared standards. Each issue raises a flag with a severity of info, warning, or critical, and contributes to a risk score from 0 to 96, which maps to a Low, Medium, or High risk band.

The routing rules

Auto-approval requires every one of the following to be true. If any check fails, the application is routed to a human assessor:

- the AI recommendation is **Auto-Approve**, and
- confidence is at or above the threshold of **0.75**, and
- the risk band is **Low**, and
- there are no critical flags.

The AI is conservative by design. It defers to a human whenever confidence is low or any safety flag is present, and novel device types or first-time manufacturers automatically lower confidence so that unfamiliar cases receive human attention.

Sample applications

The five built-in samples are designed so that two auto-approve and three route to a human, each for a clear, inspectable reason.

Device and manufacturer	Outcome	Reason
Helion HX-5, Helion Power Systems	Auto-approve	Complete, valid, low risk
Kestrel K-10, Kestrel Inverters	Auto-approve	Complete, valid, low risk
Redback RB-Hybrid 8, Redback Cells	Refer to human	Certificate near or at expiry
Voltaic VC-Wall 7, Voltaic Home Energy	Refer to human	Safety attestation not signed, a critical flag
Auralis GridNode, Auralis Grid	Refer to human	Novel device type, first-time manufacturer, low confidence

5. AI: Built-In or Live

The prototype can run its triage in two modes.

Mode	How it runs	When to use
Built-in (default)	A deterministic, rule-based triage engine inside the page. No key, works offline, fully self-contained.	Demonstrations, offline use, and anywhere reproducibility matters.
Live AI	An API key entered on the Register tab lets a language model perform the triage instead of the built-in engine.	Showing genuine model-driven triage. The key stays in the browser.

KEY HANDLING

In live mode the API key is held only in the browser for the current session. It is not stored and is not sent anywhere other than the model provider. The prototype runs fully on its built-in engine with no key.

6. Design Principles

- **The routing rules are in the application, not the model.** The confidence-and-risk routing rules are visible, testable, and stricter than the model's own recommendation.
- **Conservative by default.** The triage defers to a human whenever confidence is low or any safety flag is present.
- **Attributable throughout.** Each audit entry is attributed to a system process, a named model, or a named human, with a UTC timestamp.

- **Transparent to the assessor.** The reasoning and the routing rules behind each decision are shown in full, so the assessor can see why a case was routed the way it was.

7. Where the Decision Is Made

A production version of the registry performs the triage and the routing decision on a server, not in the browser, and not in the model. This is a deliberate design decision with three reasons behind it.

- First, integrity of the decision. If the routing logic runs in the browser, it can be inspected and altered by anyone who opens the page. Moving the risk assessment and the routing decision to a server means the outcome is produced by a controlled component that the applicant cannot reach or change. The browser submits an application and receives a decision it cannot influence.
- Second, the model does not decide. The server applies the deterministic routing rules to the model's outputs. The model contributes the risk reasoning and supporting narrative; it does not set the routing outcome. Auto-approval follows only from the rules, applied server-side, so a change in model behaviour cannot by itself change who is approved.
- Third, protection of credentials and reference data. Any model provider credential is held on the server and never exposed to the browser. Manufacturer history and other reference data that feed the rules are held server-side as well, so they cannot be read or modified from the client.

Under this design the browser is a presentation layer. It collects the application, displays the returned assessment, and provides the assessor's review workspace. The assessment, the routing decision, and the audit record are produced and held by the server.

DESIGN DECISION

Triage and routing are performed server-side. The rules produce the decision, the model produces only the reasoning, and credentials and reference data remain on the server. This keeps the routing decision reproducible, outside the model's control, and beyond the reach of the client.

8. Key Terms

Term	Meaning
Triage	The initial risk assessment of an application, producing a score, band, confidence, recommendation, and flags.
Routing rules	The transparent rules that decide whether an application is auto-approved or sent to a human.
Flag	A specific issue raised during triage, carrying a severity of info, warning, or critical, and naming the field it is based on.
Critical flag	A flag serious enough, for example an expired certificate or unsigned safety attestation, to force human review.
Confidence threshold	The minimum confidence, 0.75 , required for auto-approval.
Human-in-the-loop	A design in which a person retains authority over the final decision, with AI providing support only.
Append-only audit trail	A log to which entries are only ever added, never edited or removed, preserving a complete history.